

Hana Ahmed Sakr

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Personal Summary

Results-driven Media Engineering (computer Science) student with strong analytical and problem-solving abilities. Passionate about building scalable data pipelines, complex software systems, real-world tech solutions, analyzing bottlenecks and mastering new technologies, with an eagerness to collaborate within a fast-paced team environment.

Education

German University in Cairo (GUC) B.Sc. in Media Engineering [2023-2028]

Projects

1. **Advanced RAG Evaluation & Tuning Pipeline** | Engineered a reliable hallucination-free automated RAG pipeline in Python using ETL, achieving perfect 1.0 Context Relevance scores by optimizing source text chunking and building custom automated evaluation scripts powered by the Gemini API.
2. **Enterprise HR Management System** | Engineered an enterprise-level web application integrated with a relational SQL Server database to manage university-wide HR operations. Designed relational database schemas utilizing procedures, views, and functions to ensure secure, scalable and efficient data access.
3. **Software Engineering Student Portfolio Platform** | Engineered the front-end architecture & GUI for a scalable student portfolio platform, defining and implementing 30+ functional and non-functional user stories. Participated in project planning, task investigation, and work breakdown, utilizing GitHub to manage collaborative workflows and version control.
4. **Java Game Development: Jackaroo** | Developed a Java adaptation of the Jackaroo board game for one human player against 3 automated CPUs. Programmed complex game logic, including track navigation, collision detection and actions using Object-Oriented Programming
5. **Operating Systems Resource Manager** | Architected robust operating system logic—including system calls, process states, memory, and a custom scheduler- to optimize CPU performance and completely eliminate deadlocks. Implemented strict synchronization and concurrent resource allocation using mutexes & thread-safe logic.
6. **Full-stack Travel Web Application** | Engineered a travel platform, integrating responsive front-end GUI components with robust backend logic to ensure a seamless user experience and high system reliability. Implemented a search algorithm for destination querying and designed strict duplicate-entry prevention and error-handling mechanisms.

Technical Skills

- **Programming Languages:** Java, Python, Haskell, Prolog, JavaScript, OOP, C/C++
- **Databases:** SQL (SQL Server), Query Optimization, Supabase
- **Tools:** React, Git/GitHub, API Developer, HTML/CSS ,User Story scoping & Linux OS

Extracurricular Activities

- **Applied AI Committee Member** | IEEE GUC Student Branch
- **Associates Experience Committee Member** | TEDxGUC
- **Delegate** | GUC MUN
- **Usher** | GUC SCAD Employment Fair
- **Usher** | GUC orientation

Soft Skills

- Public Speaking & Presentation
- Team Work & Collaboration
- Time Management